



Sujit Biswas, PhD

*Senior Lecturer (Associate Professor) in Cybersecurity and FinTech
Department of Computer Science
City-St.George's, University of London, UK*

*&
Research Associate (Honorary),
Centre for Blockchain Technology (CBT),
University College London (UCL), UK*

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Google Scholar, LinkedIn*

OVERVIEW

A researcher with hands-on experience in blockchain and Artificial Intelligence. Research specialisation includes Complex **IoT applications**, IoT-generated time-series data analysis in **machine learning**, health informatics, early prediction using medical data (e.g., cancer, heart diseases, prediction), and information security-related issues in typical machine learning, **Federated Machine learning**, .

APPOINTMENTS

Senior Lecturer (Associate Professor) in Cybersecurity

08/2025 -
Cont.

City, University of London, UK

- **Teaching:** Cyber Security (PG Level), Java (UG level)
- **Research:** FinTech, IoT, AI, Blockchain, Federated Machine Learning, etc.

Lecturer (Assistant Professor) in Cybersecurity and FinTech

2024-2025

City, University of London, UK

- **Teaching:** Cyber Security and Fraud Mitigation
- **Research:** FinTech, IoT, AI, Blockchain, Federated Machine Learning, etc.

Lecturer (eq. Asst. Prof.) in Computer Science and Digital Technologies

2022- 2023

University of East London, UK

- **Teaching:** Data Structure & Algorithms, Programming (Java, Python)
Internet of Things, Artificial Intelligence, Machine Learning.
- **Research:** IoT, Blockchain, Federated Machine Learning, etc.

Research Associate (Honorary)

2022 - Cont.

Centre for Blockchain Technologies, University College London (UCL), UK

- **Research:** Blockchain and Federated Machine Learning

Research Fellow in Blockchain and AI

2021 - 2022

University of Surrey, UK

- **Research:** Blockchain and Federated Machine Learning

Assistant Professor of Computer Science and Engineering

2018 - 2021

Faridpur Engineering College, University of Dhaka, Bangladesh

- **Research:** IoT, Blockchain, Big Data, Data Analytic
- **Teaching:** Cyber Security, Data Structure & Algorithms, Structured Programming, IoT, Object Oriented Programming, Advanced Networking

Instructor (Lecturer) of Computer Engineering

2011- 2018

Jessore Polytechnic Institute, Ministry of Education, Bangladesh

- Teaching courses on Programming Languages, Digital Electronics, Networking Fundamentals

EDUCATION

PhD Computer Science and Technology*Beijing Institute of Technology (BIT), China*

Computer Science and Technology Department

Beijing Engineering Research Centre of High Volume Language Information Processing and Cloud Computing Applications

Thesis Title: Secure and Scalable Blockchain Architecture for Interoperable IoT Systems

Thesis Advisor: Prof. Dr. Li Fan (BIT), Co-Advisor: Prof. Dr. Kashif Sharif (BIT)

Master of Computer Science and Technology*Computer Science and Engineering Department*

Northwestern Polytechnical University, China

Research advisor: Prof. Wang Heipang

Bachelor of Computer Science and Information Technology*The University of Comilla, Bangladesh*

CGPA: 3.91 out of 4.00

RESEARCH PROJECTS

Duration	Details
09/2025 to 02/2028	Title: iCARE: Intelligent, Customised Admissions and Recruitment Engagement platform Budget: 268KGBP Funder: Innovate UK and Industry Role: Principal Investigator
10/2025 to 09/2028	Title: Framework for Privacy-Preserving Data Extraction and Synthesis from Multi-Domain Environments Budget: 126K GBP Funded By: Industry and City University of London. Role: Principal Investigator
2021: 1 Year	Title: Blockchain Based Public Examination Management System Budget: 0.6 million BDT Sponsor: Information and Communication Technology Division (ICTD), Bangladesh Role: Principal Investigator
2021: 1 Year	Title: Remote Management of Smart Farming Budget: 0.5 million BDT Sponsor: Information and Communication Technology Division (ICTD), Bangladesh Role: Co-Investigator

TECHNICAL SKILLS

- Simulation software: Node-red, SimuLink, WorkBench
- Programming Languages :Python, Java, C++, C, PHP, HTML, etc.
- Package Programs: *LaTeX*, MS Office, illustrator, Photoshop
- Technology :Federated Machine Learning, Machine Learning, Deep Learning, Blockchain

PROFESSIONAL MEMBERSHIP

- Fellow of the Higher Education Academy (FHEA), Advance HE, UK
- Life Member, Bangladesh Computer Society (BCS)
- Senior Member, IEEE

DISTINCTIONS AND AWARDS

- **Award:** Most Distinguish Student Award in Beijing Institute of Technology, China
Year: 2019

- **Award:** Best Teacher Award By Jashore Polytechnic Institute, Bangladesh

Year: 2011, 2012

LANGUAGE SKILLS

- English: Fluent (English was the medium of instruction of all education level)
- Chinese: Intermediate
- Bengali: Native Speaker

SYNERGISTIC ACTIVITIES

- Editorial Board Member
 - Blockchains; (2023- Continue) MDPI Journal
- Reviewer
 - IEEE Communication Surveys and Tutorials, IEEE Transaction on Industrial Informatics, Trnx on Vehicular Technology, IEEE Access, IEEE Communication Magazine, IEEE Network Magazine, IEEE Internet of Things Journals, Cyber Physical Systems, IEEE Computer Magazine, Software: Practice and Experience (Wiley, SCI), so on.
- TPC Member- Many reputed Conferences
- Supervision
 - Engineering Final Project
 - Undergraduate Thesis
 - Postgraduate Thesis (Master)

PEER-REVIEWED PAPERS [Details in [Google Scholar](#)]

Year: 2025

56. Swetha Ghanta, Prasanthi Boyapati, **Sujit Biswas**, Ashok K. Pradhan, Saraju P. Mohanty, “Enhancing privacy-preserving brain tumor classification with adaptive reputation-aware federated learning and homomorphic encryption”, *PeerJ Computer Science* , (2025).
55. AKZ Rasel Rahman, Promila Ghosh, Avi Deb Raha, **Sujit Biswas**, “Focusing on Subtleties: Class-Specific Attention-Based Deep Learning for Precise Diagnosis of Lung and Colon”, *Proceeding of the 2nd International Conference on Machine Intelligence and Emerging Technologies: MIET 2024* , (2025).
54. Ma Junchi, Hassan Nazeer Chaudhry, Farzana Kulsoom, Yang Guihua, Sajid Ullah Khan, **Sujit Biswas**, Zahid Ullah Khan, Faheem Khan, “MULTICAUSENET temporal attention for multimodal emotion cause pair extraction”, *Scientific Reports* , (2025).
53. Aritra Acharyya, Prasenjit Dey, **Sujit Biswas**, “Introduction to Real-World Applications and Implementations of IoT”, *Springer Nature Singapore* , (2025).
52. Abhiram Thiriveedhi, Swetha Ghanta, **Sujit Biswas**, Ashok K Pradhan, “ALL-Net: integrating CNN and explainable-AI for enhanced diagnosis and interpretation of acute lymphoblastic leukemia”, *PeerJ Computer Science* , (2025).
51. A Nehra, N Bansal, S Mittal, **Sujit Biswas**, RS Bali, K Naik, “A Big Data Federated Learning-based Traffic Optimization Routing Scheme for Emergency Services Provision in Autonomous Vehicles Environment”, , (2025).
50. S Chaudhary, I Budhiraja, R Chaudhary, N Kumar, **Sujit Biswas**, “Asynchronous Federated Learning Technique for Latency Reduction in STAR-RIS enabled VRCS”, , (2025).
49. Avi Deb Raha, Mrityunjoy Gain, Sudip Kumar Saha, Md Shohanur Rahman, Apurba Adhikary, Rameswar Debnath, Anupam Kumar Bairagi, **Sujit Biswas**, “Resource-Aware Monkeypox Diagnosis: Leveraging High-Capacity and Lightweight Models with Knowledge Distillation”, *IEEE* , (2025).
48. Avi Deb Raha, Mrityunjoy Gain, Sudip Kumar Saha, Md Shohanur Rahman, Apurba Adhikary, Rameswar Debnath, Anupam Kumar Bairagi, **Sujit Biswas**, “Resource-Aware Monkeypox Diagnosis: Leveraging High-Capacity and Lightweight Models with Knowledge Distillation”, *IEEE* , (2025).
47. Aritra Acharyya, Prasenjit Dey, **Sujit Biswas**, “Real-World Applications and Implementations of IoT”, *Springer Nature* , (2025).

Year: 2024

46. **Sujit Biswas**, Kashif Sharif, Zohaib Latif, Mohammed JF Alenazi, Ashok Kumar Pradhan, Anupam Kumar Bairagi, “Blockchain controlled trustworthy federated learning platform for smart homes”, *IET Communications* , (2024).
45. Md Nazmul Abdal, Md Azizul Haque, Md Mehedi Hassan, Md Nasim Adnan, Apurba Adhikary, **Sujit Biswas**, Md Shirajum Munir, Anupam Kumar Bairagi, “IoT-Based Smart Mushroom Farming, and Classification of Mushroom Using Deep Learning”, *Springer Nature Singapore* , (2024).

44. AKZ Rasel Rahman, Promila Ghosh, Avi Deb Raha, **Sujit Biswas**, Anupam Kumar Bairagi, “Focusing on Subtleties: Class-Specific Attention-Based Deep Learning for Precise Diagnosis of Lung and Colon Cancers”, *Springer Nature Singapore* , (2024).
43. Sanjana Singamsetty, Swetha Ghanta, **Sujit Biswas**, Ashok Pradhan, “Enhancing machine learning-based forecasting of chronic renal disease with explainable AI”, *PeerJ Computer Science* , (2024).
42. Fatema Tuz Zohora, **Sujit Biswas**, Anupam Kumar Bairagi, Kashif Sharif, “Nature-based Bengali Picture Captioning using Global Attention with GRU”, *2024 IEEE 34th International Workshop on Machine Learning for Signal Processing (MLSP)* , (2024).
41. Sadaf Bukhari, Kashif Sharif, Liehuang Zhu, Chang Xu, Fan Li, **Sujit Biswas**, “Dynamic Fine-grained SLA Management for 6G eMBB-plus Slice using mDNN & Smart Contracts”, *IEEE Transactions on Services Computing* , (2024).
40. Sadaf Bukhari, Kashif Sharif, Liehuang Zhu, Chang Xu, Fan Li, **Sujit Biswas**, “CIC-SIoT: Clean-Slate Information-Centric Software-Defined Content Discovery and Distribution for Internet-of-Things”, *IEEE Internet of Things Journal* , (2024).
39. Prakhar Consul, Neeraj Joshi, Ishan Budhiraja, **Sujit Biswas**, Neeraj Kumar, Sachin Sharma, Ajith Abraham, “A Reliable Zero-Trust Network for Task Offloading in Vehicular Systems Using an Asynchronous Federated Learning Approach in 6G”, *Proceedings of the SIGCOMM Workshop on Zero Trust Architecture for Next Generation Communications* , (2024).
38. Anindya Nag, Hirak Mondal, Md Mehedi Hassan, Taher Al-Shehari, Mohammed Kadrie, Muna Al-Razgan, Taha Alfakih, **Sujit Biswas**, Anupam Kumar Bairagi, “TumorGANet: A transfer learning and generative adversarial network-Based data augmentation model for brain tumor classification”, *IEEE access* , (2024).
37. Liehuang Zhu, Sadaf Bukhari, Kashif Sharif, Fan Li, Shumaila Fardous, **Sujit Biswas**, “A Novel Merging Framework for Homogeneous and Heterogeneous Blockchain Systems”, *International Conference on Wireless Artificial Intelligent Computing Systems and Applications* , (2024).
36. Fatema Jannat Dihan, Saydul Akbar Murad, Abu Jafar Md Muzahid, KM Uddin, Mohammed JF Alenazi, Anupam Kumar Bairagi, **Sujit Biswas**, “MpoxSLDNet: A Novel CNN Model for Detecting Monkeypox Lesions and Performance Comparison with Pre-trained Models”, , (2024).
35. Sajib Chatterjee, Rameswar Debnath, **Sujit Biswas**, Anupam Kumar Bairagi, “Prediction of RNA Secondary Structure Using Butterfly Optimization Algorithm”, *Human-Centric Intelligent Systems* , (2024).
34. Zhaocheng Wang, Xueying Liu, Xinming Shao, Abdullah Alghamdi, Mesfer Alrizq, Md Shirajum Munir, **Sujit Biswas**, “An Optimized and Scalable Blockchain-Based Distributed Learning Platform for Consumer IoT”, **Mathematics**, (2023).

Year: 2023

33. Suranjeet Chowdhury Avik, **Sujit Biswas**, Md Atiqur Rahaman Ahad, Zohaib Latif, Abdullah Alghamdi, Hamad Abosaq, Anupam Kumar Bairagi, “Challenges in Blockchain as a Solution for

- IoT Ecosystem Threats and Access Control: A Survey”, **arXiv preprint arXiv:2311.15290**, (2023).
32. Suranjeet Chowdhury Avik, **Sujit Biswas**, Md Atiqur Rahaman Ahad, Zohaib Latif, Abdullah Alghamdi, Hamad Abosag, Anupam Kumar Bairagi, “Challenges in Blockchain as a Solution for IoT Ecosystem Threats and Access Control: A Survey”, **arXiv e-prints**, (2023).
 31. Muhammad Sohail,Zohaib Latif, Shahzeb Javed,**Sujit Biswas**, Sahar Ajmal, Umer Iqbal, Mohsin Raza, “Routing protocols in vehicular adhoc networks (vanets): A comprehensive survey” , , 2023 ().
 30. Nusrat Jahan Prottasha, Saydul Akbar Murad, Abu Jafar Md Muzahid, Masud Rana, Md Kowsher, Apurba Adhikary, **Sujit Biswas**, Anupam Kumar Bairagi , “Impact learning: A learning method from feature’s impact and competition” , *Journal of Computational Science* **SCI-IF:2.69**, (2023).
 29. **Sujit Biswas**, Zigang Yao, Lin Yan, Abdulmajeed Alqhatani, Anupam Kumar Bairagi, Fatima Asiri, Mehedi Masud, “Interoperability benefits and challenges in smart city services: Blockchain as a solution” , **Electronics**, (2023).
 28. **Sujit Biswas**, Z Yao, L Yan, A Alqhatani, AK Bairagi, F Asiri, M Masud, “Interoperability Benefits and Challenges in Smart City Services: Blockchain As a Solution. Electronics 2023, 12, 1036” , , (2023).
 27. Zohaib Latif, Qasim Umer, Choonhwa Lee, Kashif Sharif, Fan Li, and **Sujit Biswas**, “A Machine Learning-Based Anomaly Prediction Service for Software-Defined Networks” , **Sensors**, SCI-IF:3.847 (2022).
 26. Abdullah Alghamdi,Jiang Zhu, Guocai Yin, Mohammad Shorfuzzaman,Nawal Alsufyani,Sultan Alyami,and **Sujit Biswas**, “Blockchain Empowered Federated Learning Ecosystem for Securing Consumer IoT Features Analysis” , **Sensors**, SCI-IF:3.847 (2022).
 25. Jun Wang,Jiang Zhu, Minghui Zhang, Iqbal Alam, **Sujit Biswas**, “Function Virtualization Can Play a Great Role in Blockchain Consensus” , **IEEE Access**, SCI-IF:3.367 (2022).
 24. Yiyang Liu, Guangxing Shan, Yucheng Liu, Abdullah Alghami, Iqbal Alam, **Sujit Biswas**, “Blockchain Bridges Critical National Infrastructures: E-healthcare Data Migration Perspective” , *IEEE Access* **SCIE-IF 3.367** , 2022 (<https://ieeexplore.ieee.org/abstract/document/9727100>).
 23. Puspendu Biswas Paul, **Sujit Biswas**, Anupam Kumar Bairagi , Mehedi Masud, “Data-Driven Decision Making for Smart Cultivation” , **7th IEEE International Symposium on Smart Electronic Systems (iSES)**, (2021).
 22. Sumon Reza, **Sujit Biswas**,Abdullah Alghamdi, Mesfer Alrizq, Anupam Kumar Bairagi , Mehedi Masud, “ACC: Blockchain Based Trusted Management of Academic Credentials” , **7th IEEE International Symposium on Smart Electronic Systems (iSES)**, (2021).
 21. **Sujit Biswas**, Kashif Sharif, Fan Li, Anupam Kumar Bairagi, Zohaib Latif, Saraju P Mohanty, “Globechain: An interoperable blockchain for global sharing of healthcare data—a covid-19 perspective” , **IEEE Consumer Electronics Magazine**, (2021).

20. Zohaib Latif, Kashif Sharif, Fan Li, Md Monjurul Karim, **Sujit Biswas**, Madiha Shahzad, “Member, IEEE, and Saraju P” , **Mohanty, “DOLPHIN: Dynamically Optimized and Load Balanced Path for Inter-domain SDN Communication,” IEEE Transactions On Network And Service Management**, (2021).
19. Zohaib Latif, Kashif Sharif, Fan Li, Md Monjurul Karim, **Sujit Biswas**, Madiha Shahzad, Saraju P Mohanty, “DOLPHIN: Dynamically optimized and load balanced path for inter-domain SDN communication”, **IEEE Transactions on Network and Service Management**, (2020).
18. Anupam Kumar Bairagi, Mehedi Masud, Md Shirajum Munir, Abdullah-Al Nahid, Sarder Fakhrul Abedin, Kazi Masudul Alam, **Sujit Biswas**, Sultan S Alshamrani, Zhu Han, Choong Seon Hong, “Controlling the outbreak of COVID-19: A noncooperative game perspective”, **Ieee Access**, (2020).
17. **Sujit Biswas**, Kashif Sharif, Fan Li, Iqbal Alam, Saraju Mohanty, “DAAC: Digital Asset Access Control in a Unified Blockchain Based E-Health System”, **IEEE Transactions on Big Data**, (2020).
16. **Sujit Biswas**, Kashif Sharif, Fan Li, Saraju Mohanty, “Blockchain for e-health-care systems: Easier said than done”, **Computer**, (2020).
15. **Sujit Biswas**, Kashif Sharif, and Fan Li, Zohib Latif, and Sarju Mohanty , “Interoperability and Synchronization Management of Blockchain-Based Decentralized e-Health Systems”, **IEEE Transactions on Engineering Management, SCIE-IF:2.784** (2020).
14. Anupam Kumar Bairagi, Mehedi Masud, Do Hyeon Kim, Md Shirajum Munir, Abdullah Al Nahid, Sarder Fakhrul Abedin, Kazi Masudul Alam, **Sujit Biswas**, Sultan S Alshamrani, Zhu Han, Choong Seon Hong, “A noncooperative game analysis for controlling covid-19 outbreak”, **ACM Computing Surveys**, (2020).
13. Iqbal Alam, Kashif Sharif, Fan Li, Zohaib Latif, Md Monjurul Karim, **Sujit Biswas**, Boubakr Nour, and Yu Wang, “A Survey of Network Virtualization Techniques for Internet of Things Using SDN and NFV”, **ACM Computing Surveys, SCIE-IF(5.55)** (2020).
12. Zohib Latif, Kashif Sharif, Fan Li, Monjurul Karim, **Sujit Biswas**, Yu Wang, “A Comprehensive Survey of Interface Protocols for Software Defined Networks”, **Elsevier: Journal of Computer Network and Applications (JNCA), SCIE-IF:5.273** (2020).
11. **Sujit Biswas**, Kashif Sharif, Fan Li, Sabita Maharjan, Saraju P. Mohanty, Yu Wang, “PoBT: A Light Weight Consensus Algorithm for Scalable IoT Business Blockchain”, **IEEE Internet of Things Journal**, (2019).
10. Boubakr Nour, Kashif Sharif, Fan Li, **Sujit Biswas**, Hassine Moun gla, Mohsen Guizani, Yu Wang, “A survey of Internet of Things communication using ICN: A use case perspective”, **Computer Communications**, 2019 (**IF:11.705**).
9. Iqbal Alam, Kashif Sharif, Fan Li, Zohaib Latif, Md Monjurul Karim, Boubakr Nour, **Sujit Biswas**, Yu Wang, “IoT Virtualization: A Survey of Software Definition & Function Virtualization Techniques for Internet of Things”, **arXiv preprint arXiv: 1902.10910**, (2019).

8. Alam, K Sharif, F Li, Z Latif, MM Karim, B Nour, **Sujit Biswas**, Y Wang, , “IoT virtualization: A survey of software definition function virtualization techniques for internet of things. arXiv 2019”, **arXiv preprint arXiv: 1902.10910**, (2019).
7. **Sujit Biswas**, Kashif Sharif, and Fan Li, Boubakar Nour, Yu Wang, “ A Scalable Blockchain Framework for Secure Transactions in IoT”, **IEEE Internet of Things Journal**, 2018 (**IF:11.705**).
6. **Sujit Biswas**, Kashif Sharif, and Fan Li, Yang Liu, “3P Framework: Customizable Permission Architecture for Mobile Applications”, *The 12th International Conference on Wireless Algorithms, Systems, and Applications (WASA 2017)* **Springer Lecture Notes**, vol.10251 (2017).
5. **Sujit Biswas**, Wang Haipeng and Javed Rashid, “Android Permissions Management at App Installing”, **International Journal of Security and Its Applications**, 3 (PP. 223-232,(2016)).
4. **Sujit Biswas**, “Mobile Banking Based a New E-commerce Architecture: Developing Countries Perspective”, *International Journal of u- and e-Service, Science and Technology* **Vol.8, No. 2**, 2015 (P.P 221-236).
3. **Sujit Biswas**, “Mobile Banking Based a New E-commerce Architecture: Developing Countries Perspective”, *International Journal of u- and e-Service, Science and Technology* **Vol.8, No. 2**, 2015 (P.P 221-236).
2. **Sujit Biswas**, “GSM Verification Based Secure E-Voting Framework”, *International Journal of u- and e-Service, Science and Technology* **Vol.8, No. 1**, 2015 (pp. 231-238).
1. **Sujit Biswas**, “DOLPHIN: Dynamically Optimized and Load Balanced Path for Inter-Domain SDN Communication”, , ().

Relationship	Referees
Line Manager	Professor Golnaz Badkobeh Head of Computer Science Department City St. George’s, University of London, UK golnaz.badkobeh@city.ac.uk
PhD Supervisor	Dr. Fan Li Professor School of Computer Science Beijing Institute of Technology, China fli@bit.edu.cn
Collaborator	Professor Muttukrishnan Rajarajan Professor City St. George’s, University of London, UK r.muttukrishnan@city.ac.uk

(Sujit Biswas)
21 September 2025